HK IMO Training 2023-2024
Problem Set 6
For 11 November 2023

Problem 21. Let $ABCD$ be a trapezoid inscribed in a circle with centre O, where AB and DC are parallel. The lines AD and BC meet at E. A circle passing through E and O meets the segments BC and AD at P and Q respectively. Prove that $BP = DQ$.

Problem 22. For any positive integer n with k digits, we say that a positive integer m is a *weak divisor* of n if m divides the number obtained by removing the leftmost a digits and the rightmost b digits of n, where a and b are nonnegative integers with $a + b < k$. For example, 19 is a weak divisor of 13579 because 19 divides 57, while 101 is a weak divisor of 2023 because 101 divides 202. Find all positive integers d such that there are finitely many positive integers n for which d is not a weak divisor of n.

Problem 23. Let a, b, c be real numbers satisfying $b > 2c > 4a$ and

$$2(a^3 + b^3 + c^3) + 15(ab^2 + bc^2 + ca^2) > 16(a^2b + b^2c + c^2a) + 2abc.$$

Prove that $4a + b > 4c$.

Problem 24. Let n be a positive integer. There are $2n - 1$ coins lying in a row. Initially, only the middle coin (the nth coin) has its head facing upwards, while all other coins have their tails facing upwards. In each step, we can choose at least three consecutive coins such that the leftmost and the rightmost coins have their tails facing upwards and all other coins have their heads facing upwards, and we flip all these coins. At most how many steps can be performed?